Detailed Technical Audit: Research Paper + Survey Paper

IEEE-oriented scientific review, consistency audit, statistical validation audit, figure/table audit, and correction plan

Files audited: “Hybrid E-com AI Survey Paper (Formatted)” (17 pages) and “Launch-Blind Product Viability” (21 pages). The audit uses the supplied documents as the primary source of truth and adds a limited external verification pass for IEEE requirements and selected bibliography items.

Overall conclusion: the research paper contains a substantially stronger and more defensible methodology than the current survey paper describes. The research paper is not yet submission-ready because several numerical/figure inconsistencies, provenance gaps, evaluation-definition gaps, and IEEE-format issues remain. The survey should be revised *after* the research paper is frozen, using the research paper's final methodology and results as the source of truth.

1. Executive decision

Area	Current status	Priority	Decision
Research concept	Strong and coherent: historical review learning + specification-to-aspect bridging + launch-blind prediction.	—	Keep
Research paper methodology	Defensible core, but several protocol details are under-specified.	P0/P1	Fix before submission
Research paper results	Useful external holdouts, but Figure 6 conflicts with the text and some metrics need clearer definitions/CIs.	P0	Recompute/regenerate
Research paper figures	Several figures contain stale sample size or over-interpretive labels; Figure 3's verdict is not supported by all metrics.	P0/P1	Regenerate
Survey methodology	Currently describes an older, simpler LLM→XGBoost system rather than the actual launch-blind research system.	P0	Rewrite proposed-method section
Survey results comparison	Figure 1 mixes accuracy and AUC from different tasks and is misleading.	P0	Replace
References	Survey numbering is not IEEE citation order; selected metadata/citation descriptions are wrong or need verification.	P0/P1	Renumber and verify
IEEE formatting	Neither document is safely treated as final IEEE submission formatting; research paper also lacks a standard affiliation block.	P0	Move into target IEEE template

Key point: do not throw away the research paper. The main architecture is worth retaining. The immediate goal is to make the evidence, data provenance, figures, and claims line up exactly.

2. Highest-priority findings

ID	Finding	Paper	Severity	What to do
R-01	Figure 6 reports ρ≈0.437 and AUC=0.826, while the text reports ρ=0.529 and AUC=0.903 for the same holdout.	Research	P0	Recompute from frozen predictions; regenerate figure and all dependent text.
R-02	Abstract CI [0.31,0.61] conflicts with validation CI [0.377,0.654].	Research	P0	Use the exact bootstrap output everywhere.
R-03	Figure 1/3 artwork says n=732, while the cleaned dataset and captions say n=631.	Research	P0	Regenerate figures from the final 631-product analysis object.
R-04	Figure 3's “BEST/Good/Weak” verdict overstates XGBoost's overall dominance; Gradient Boosting wins several classification-derived metrics and ties AUC.	Research	P0	Remove verdict column or predefine a primary metric hierarchy.
R-05	The final launch-blind pipeline depends on category profiles, but it is not explicit enough that every profile component is training-only in each holdout.	Research	P0	Add a feature-provenance/leakage table and audit every input.
R-06	Specification/launch-date acquisition is not sufficiently reproducible: the paper does not say exactly where launch-day specifications came from or how launch dates were established.	Research	P0	Document source, retrieval/archive method, date, fields, and availability rules.
R-07	Brand-recognition tier is used in the headline model but its construction and launch-time availability are not defined.	Research	P0	Define it and prove it uses only pre-launch information; otherwise remove it.

ID	Finding	Paper	Severity	What to do
S-01	Survey describes a different methodology and an inconsistent dataset (43.6k/640/5 vs 74k/1,051/Amazon+Google Play).	Survey	P0	Rewrite the proposed method and dataset from the research paper.
S-02	Survey Figure 1 compares literature classification accuracy against proposed AUC.	Survey	P0	Separate tasks/metrics; do not put them on one performance ranking bar chart.
S-03	Survey Table 2 contains “Target R^2>0.90” and “TBD” instead of final results.	Survey	P0	Replace with actual validated results or remove the table row.
S-04	Survey reference numbering is not in IEEE first-citation order and has large numbering gaps.	Survey	P0	Renumber all citations and references after content is frozen.

3. Research paper — detailed audit

Source of truth used for this section: the supplied 21-page research paper. Page numbers below refer to that PDF.

R-01 — P0 — Research

Location: p.13, Fig. 6

Issue: The plotted holdout statistic is ρ=0.437 with AUC=0.826 and CI [0.28,0.58], while the surrounding text, Table 2, abstract, and conclusion use ρ=0.529 and AUC=0.903.

Why it matters: This is not a cosmetic difference. A reviewer cannot know which result is the actual experiment.

Required correction: Freeze the holdout prediction file and regenerate the scatter plot from the exact same 118 predictions used for the reported statistics. Then update abstract, Table 2, text, caption, and conclusion from that single result object.

Verification / evidence needed: Also verify whether the figure was generated from an earlier model version. Do not manually edit the annotation; regenerate it.

R-02 — P0 — Research

Location: p.1 vs p.12

Issue: The abstract gives 95% CI [0.31,0.61], while the validation section gives bootstrap 95% CI [0.377,0.654].

Why it matters: Conflicting uncertainty intervals are a direct reproducibility problem.

Required correction: Use one exact bootstrap procedure and report its exact interval everywhere.

Verification / evidence needed: Store the random seed, bootstrap method, resampling unit, number of resamples, and statistic used.

R-03 — P0 — Research

Location: p.9–10, Fig. 1 and Fig. 3

Issue: The chart artwork states n=732 even though the final cleaned dataset is 631 products and the captions/table text state n=631.

Why it matters: This creates an immediate credibility problem and suggests figures came from an earlier experiment.

Required correction: Regenerate all model-comparison figures from the final 631-row analysis dataset.

Verification / evidence needed: Search every figure/table caption and embedded image for 640, 631, 732, 118, 125, 505, 513 and reconcile each number.

R-04 — P0 — Research

Location: p.10, Fig. 3

Issue: The “BEST/Good/Weak” verdict implies an overall ranking, but Gradient Boosting has higher accuracy, precision, recall, specificity, F1 and ties AUC; XGBoost leads the primary regression metrics and ties AUC.

Why it matters: The current verdict is not a neutral statistical summary and is not supported by a predeclared decision rule.

Required correction: Remove the verdict column. State a predeclared selection rule such as: R^2 is the primary regression metric, RMSE secondary, AUC an auxiliary extreme-quintile diagnostic, runtime tertiary.

Verification / evidence needed: If you want a model-selection score, define it before evaluation and report how it is calculated.

R-05 — P0 — Research

Location: p.7, p.10–14

Issue: The paper says category priors are training-only in the random holdout, but the bridging prompt uses price quartiles, mean aspect scores, pain points and strengths. It is not explicit that every one of these profile elements is computed only from training data for every holdout.

Why it matters: Any post-launch or test-product information entering the category profile would violate the launch-blind claim.

Required correction: Add a “feature provenance and leakage guard” table listing every serving input, its source, timestamp, and whether it is computed from training-only data.

Verification / evidence needed: For both holdouts, verify category statistics, pain points, strengths, calibration reference distributions, and trust weights are frozen before test predictions.

R-06 — P0 — Research

Location: p.12, p.18

Issue: The paper says the bridge sees “launch-day spec sheets and marketing copy” but does not give a reproducible acquisition protocol for those specifications or launch dates.

Why it matters: The entire launch-blind claim depends on what was actually available on launch day.

Required correction: Document the exact source of each specification, how launch date was obtained, how pages were archived/recovered, and what fields were retained.

Verification / evidence needed: Add a small appendix/data card describing the specification source and date-provenance rules. If launch date is proxied by first review date, say so explicitly.

R-07 — P0 — Research

Location: p.7, p.17

Issue: Brand-recognition tier is a headline feature, but the construction is not defined. The application also conditions on company/product maturity, which is not clearly mapped to the training feature list.

Why it matters: An undefined feature can hide leakage or make the experiment irreproducible.

Required correction: Define exact bins, data source, cutoff date, and mapping from brand/product maturity to the model feature. If the tier uses post-launch popularity, remove it from the launch-blind model.

Verification / evidence needed: Run a no-brand-tier ablation and report the effect on both holdouts.

R-08 — P0 — Research

Location: p.6, p.12

Issue: The success label is min–max normalized within category. The paper does not explicitly state whether category min/max parameters for evaluation targets and training labels are fitted only on the training split.

Why it matters: For pooled holdout Spearman, using test-set extrema to normalize the target can leak test distribution information and can change between-category ordering.

Required correction: Fit label-scaling parameters on training data only for each evaluation split; apply them to the test outcomes. For within-category Spearman/AUC, the issue is less consequential, but pooled metrics must use a clean protocol.

Verification / evidence needed: Recompute pooled metrics under the training-only scaling rule and compare with current results.

R-09 — P0 — Research

Location: p.13

Issue: Temporal split uses 505 training + 125 test = 630 products, while the cleaned dataset contains 631.

Why it matters: One product is unaccounted for, which invites questions about filtering, missing launch date, or exclusion.

Required correction: State exactly why one product is omitted and include it in the sample-flow diagram.

Verification / evidence needed: Add a temporal split table with total eligible, excluded, training, validation/CV, and test counts.

R-10 — P0 — Research

Location: p.6–7

Issue: The paper says the LLM sees star rating as context but is “banned as a source of scores.” Because the target contains rating, the model can still use the rating implicitly.

Why it matters: This is a subtle target-proxy concern: LLM-generated aspect scores may encode the star rating even if the prompt says not to.

Required correction: Run a controlled extraction ablation with rating hidden from the LLM. Compare aspect-score correlation, model performance, and score distributions. If the difference is material, use the rating-blinded extraction for the headline model.

Verification / evidence needed: Add a human/automatic check for rating leakage into aspect outputs.

R-11 — P1 — Research

Location: p.6

Issue: Only 30 reviews were hand-validated, checking null behavior and evidence phrase presence.

Why it matters: Thirty examples are a useful engineering gate but weak evidence for the quality of 43,000+ automated annotations.

Required correction: Call it a “smoke-test/format gate,” not a validation of ABSA quality. Add a stratified manual annotation study on a substantially larger sample with at least two annotators.

Verification / evidence needed: Report aspect presence precision/recall/F1 and score agreement (e.g., MAE or correlation), plus inter-annotator agreement.

R-12 — P1 — Research

Location: p.6

Issue: The 0–10 aspect score scale is not accompanied by a detailed scoring rubric.

Why it matters: Without anchors, an 8 vs 6 is model-dependent and difficult to reproduce across providers.

Required correction: Include score anchors (e.g., what 0, 2, 5, 8, 10 mean) and one or two examples per aspect in an appendix.

Verification / evidence needed: Evaluate score stability across providers and repeated calls.

R-13 — P1 — Research

Location: p.6

Issue: Multi-provider extraction uses Gemini Flash, Llama-3.3-70B, Cerebras, Mistral and SambaNova, but exact model IDs/versions are not consistently specified.

Why it matters: Hosted model aliases can change; reproducibility requires exact model/version and date.

Required correction: Record exact provider + model ID + API version + temperature + max tokens + schema version + date.

Verification / evidence needed: Add a provider audit table: valid JSON rate, skipped rate, latency, cost if available, and agreement.

R-14 — P1 — Research

Location: p.7

Issue: The paper states monotonic constraints improve out-of-sample generalization, but no unconstrained-vs-constrained result is shown.

Why it matters: That is an empirical claim requiring evidence.

Required correction: Add a paired ablation: unconstrained XGBoost vs monotone-constrained XGBoost on identical splits.

Verification / evidence needed: Report CV R^2/RMSE and both external holdouts. Also report whether the constraint changes category-level behavior.

R-15 — P1 — Research

Location: p.7

Issue: Price is constrained to be non-increasing and brand tier/aspects to be non-decreasing.

Why it matters: These are modeling assumptions, not discovered causal effects.

Required correction: Phrase them as constraints: “the model was constrained so…” rather than “viability falls…” as an empirical finding.

Verification / evidence needed: Run an unconstrained sensitivity analysis and explain the domain rationale for each sign.

R-16 — P1 — Research

Location: p.14

Issue: Price-free ablation reports r=0.147 with price percentile and calls the signal “largely orthogonal.”

Why it matters: r=0.147 is better described as weakly correlated; no confidence interval is supplied.

Required correction: Report Pearson or Spearman explicitly, with CI, and use “weakly correlated” unless a formal orthogonality criterion is defined.

Verification / evidence needed: Add a scatter/rank plot only if it helps; otherwise a small ablation table is enough.

R-17 — P1 — Research

Location: p.12–14

Issue: The primary external metric is pooled Spearman, but the paper itself says pooled values benefit from between-category structure; within-category temporal ρ falls to 0.16.

Why it matters: A reviewer may view 0.590 pooled as an optimistic headline if the practical question is ranking products within their category.

Required correction: Report per-category ρ and AUC, plus macro-average and pooled metrics. Make the temporal within-category result prominent rather than burying it.

Verification / evidence needed: Add a category-by-category validation table with n, Spearman ρ, CI, AUC, and failure/abstention rate.

R-18 — P1 — Research

Location: p.12–14

Issue: AUC is computed for top-vs-bottom quintiles, not the whole product population. Confusion-matrix totals also need protocol clarification.

Why it matters: Extreme-quintile classification is a useful diagnostic but is not general binary classification accuracy.

Required correction: Rename it consistently to “top-vs-bottom quintile AUC.” Explicitly state that the middle 60% is excluded. Define whether quintiles are fixed globally or recomputed within each fold.

Verification / evidence needed: Recompute pooled confusion matrices from out-of-fold predictions and verify that sample counts match the stated labeling rule.

R-19 — P1 — Research

Location: p.10

Issue: Figure 4 uses “flop” and “hit” labels.

Why it matters: The paper defines a flop as bottom-quintile survivor, not a product that sold poorly in an absolute sense.

Required correction: Use “bottom-quintile” and “top-quintile” throughout figures/tables unless you explicitly define the shorthand.

Verification / evidence needed: This also aligns better with the survivorship limitation.

R-20 — P1 — Research

Location: p.7–9

Issue: Runtime comparisons are based on one reported run and lack hardware/software details.

Why it matters: Training time can vary substantially by CPU/GPU, BLAS, thread count, and implementation.

Required correction: Either remove runtime as a model-selection argument or report hardware, library versions, thread count, repeats and mean±SD.

Verification / evidence needed: If runtime is kept, benchmark all methods under the same environment.

R-21 — P1 — Research

Location: p.12–15

Issue: AUC CIs are not reported for the two primary holdouts, despite small extreme-class samples.

Why it matters: AUC can be unstable when the top/bottom groups are small.

Required correction: Add bootstrap CIs for AUC and, ideally, per-category AUC CIs.

Verification / evidence needed: Report the number of positives/negatives used for each AUC.

R-22 — P1 — Research

Location: p.15

Issue: Trust tiers (STRONG/MODERATE/WEAK) appear to be assigned after observing correlations, but thresholds are not defined before validation.

Why it matters: Post-hoc tier boundaries can overfit the same validation set they summarize.

Required correction: Either predefine thresholds using training data or avoid categorical tiers and show continuous fidelity with uncertainty.

Verification / evidence needed: If tiers remain, freeze the thresholds before the final holdout and report them in Method.

R-23 — P1 — Research

Location: p.10–18

Issue: The “confidence” and abstention mechanism is described, but there is no calibration/coverage evaluation.

Why it matters: A weighted score is not automatically a calibrated confidence estimate.

Required correction: Add a risk–coverage curve: as the system abstains on low-confidence cases, how does error change? Report coverage at several thresholds and calibration between predicted confidence and observed error.

Verification / evidence needed: Use training-only data to set thresholds; evaluate once on the external holdout.

R-24 — P1 — Research

Location: p.10, p.17

Issue: The 0–100 rank-calibrated viability range and percentile reference distribution are not fully specified.

Why it matters: If the reference distribution contains test products, calibration leaks test information.

Required correction: Define the reference set and fit it on training data only for each split.

Verification / evidence needed: Add a calibration diagram or at least a table showing coverage/error of the reported ranges.

R-25 — P1 — Research

Location: p.6

Issue: The 0.40/0.35/0.25 target weights are design choices with no sensitivity analysis.

Why it matters: The model is trained to a constructed target; conclusions may depend on those weights.

Required correction: Run label-weight sensitivity analyses: equal weights and several plausible alternatives. Report how Spearman/AUC and product ordering change.

Verification / evidence needed: State clearly that the target is a review-derived viability proxy, not sales.

R-26 — P1 — Research

Location: p.6

Issue: Review velocity is not precisely defined in terms of observation window and product age.

Why it matters: A product with different exposure time can have a different velocity statistic; the target may partly reflect observation-window choices.

Required correction: Define velocity mathematically and use a common post-launch observation window if possible, or justify the exposure normalization.

Verification / evidence needed: Run a sensitivity analysis using a fixed review window.

R-27 — P1 — Research

Location: p.5–6

Issue: The fraud filter thresholds (>40% in 14 days, >30% near duplicates, Jaccard >0.7, J-shaped ratings) are heuristic and not validated.

Why it matters: The filter can remove legitimate launches or fail sophisticated manipulation.

Required correction: Add a short rationale or sensitivity analysis showing results with and without the filter.

Verification / evidence needed: Report the nine removed products by category and the exact exclusion reason counts.

R-28 — P1 — Research

Location: p.18

Issue: Survivorship bias is acknowledged, but no sensitivity analysis is possible because products with <5 reviews are absent.

Why it matters: This limitation directly constrains what “success” means.

Required correction: Keep the limitation, but avoid language implying the model can detect absolute flops. Consider future external data with sales/rank information.

Verification / evidence needed: Use “survivor-product viability ranking” where precision is important.

R-29 — P1 — Research

Location: p.17

Issue: The spec coach asks questions “ordered by how much each answer would sharpen the prediction,” but no experiment proves that ordering.

Why it matters: This is a product feature claim, not a validated research result.

Required correction: Either evaluate the coach with a small user study/ablation or describe it as an implementation heuristic.

Verification / evidence needed: Do not present it as experimentally validated unless tested.

R-30 — P2 — Research

Location: throughout

Issue: Several phrases are rhetorical rather than scientific: “only honest thing,” “default to beat,” “not in dispute,” “answers the most common objection,” “oracle they should not trust.”

Why it matters: They make the paper sound argumentative and can distract from the empirical contribution.

Required correction: Replace with neutral, testable language.

Verification / evidence needed: Keep the strong limitations; remove rhetorical framing.

4. Research paper — evaluation redesign

The current evaluation is promising, but the following additions would make it much harder for a reviewer to challenge the central claim.

Analysis	Current paper	Recommended final analysis	Why it matters
Main launch-blind holdout	n=118; pooled ρ/AUC	Recompute exact predictions; report ρ + 95% CI, AUC + 95% CI, MAE/RMSE if score is continuous	Primary external evidence
Temporal holdout	n=125; pooled ρ/AUC	Same metrics + per-category and macro-average metrics	Shows whether signal survives time/category shift
Per-category performance	Only mean within-category ρ	Five-category table with n, ρ, CI, AUC, CI, RMSE	Reveals where method actually works
Price baseline	ρ=.386	Add category-mean baseline, price-only model, random/rank baseline, and price-free proposed model	Shows incremental value beyond price
Model ablation	Aspects-only R^2=.566; price/brand effects partly shown	Full factorial ablation: aspects; +price; +brand; +category; constrained vs unconstrained	Quantifies each design choice
LLM extraction validation	30-review gate	Human annotation study, inter-rater agreement, aspect detection F1, score MAE/correlation	Validates the measurement layer
Rating-blind extraction	Not tested	Repeat ABSA without star-rating input	Tests hidden target-proxy leakage
Confidence/abstention	Mechanism described	Risk-coverage curve + calibration/coverage table	Turns “confidence” into measured reliability
Target sensitivity	One 40/35/25 label	Alternative weightings + rank stability	Tests dependence on arbitrary label engineering
Specification bridge	Aspect r values	Per-aspect calibration, category-specific bridge error, provider sensitivity	Shows whether the bridge is reliable

Recommended primary result table

Model / evaluation	Spearman ρ	95% CI	AUC (top/bottom quintile)	AUC 95% CI	Within-category ρ
Leakage-controlled holdout (n=118)	0.529*	RECOMPUTE / freeze	0.903*	ADD	0.39*
Forward temporal holdout (n=125)	0.590	[0.486,0.696]	0.942	ADD	0.16
Price-only baseline	0.386	ADD	ADD	ADD	ADD
Price-free proposed	0.475	ADD	ADD	ADD	ADD

*Use the exact final recomputed values. Do not retain conflicting figure annotations.

Recommended ablation table

Variant	Launch-blind inputs	Purpose
A0	Category prior only	Lower-bound / naive baseline
A1	Price only	Measures price signal
A2	LLM aspects only	Measures semantic signal
A3	Aspects + price	Headline signal without brand
A4	Aspects + price + brand tier	Tests brand contribution
A5	Aspects + price + brand + category	Full headline feature set
A6	A5, unconstrained	Tests whether monotonic constraints help
A7	A5, rating-blind extraction	Tests hidden rating leakage

Recommended new figures

Figure	What it should show	Priority
New Fig. 1	Final launch-blind architecture: historical review learning → bridge → XGBoost → confidence/abstention.	P0
New Fig. 2	Holdout predicted vs actual viability, regenerated from final 118 predictions with exact ρ/AUC annotation.	P0
New Fig. 3	Temporal holdout: predicted vs actual or rank-vs-rank, with ρ/AUC and CI.	P1
New Fig. 4	Per-category Spearman ρ with 95% CIs for random and temporal holdouts.	P1
New Fig. 5	Risk–coverage curve for confidence/abstention.	P1
New Fig. 6	Unconstrained vs monotone-constrained performance across the same splits.	P1

Figure	What it should show	Priority
New Fig. 7	Per-aspect bridging fidelity with CIs, plus sample counts.	P1

5. Survey paper — detailed audit

The survey should be treated as a literature-and-methodology companion to the research paper. It must not describe an older version of the system.

S-01 — P0 — Survey

Location: p.1, Abstract

Issue: The abstract says approximately 43,600 reviews / 640 products / five hardware categories, while Section 5 says 74,000+ reviews / 1,051 products including Amazon and Google Play.

Why it matters: The survey contradicts itself and the current research paper.

Required correction: Use the research paper's final dataset: 43,000+ reviews, 640 initial products, five physical categories; 631 after fraud filtering for model development.

Verification / evidence needed: If a separate exploratory dataset really exists, label it separately and explain why it is not the final experiment.

S-02 — P0 — Survey

Location: p.1, p.10–12

Issue: The proposed system is described as direct LLM ABSA + XGBoost on review features, but the actual research system is launch-blind: historical reviews train the aspect-success relationship, then a specification-to-aspect bridge supplies the new product's features.

Why it matters: This is the central methodological mismatch between the papers.

Required correction: Rewrite the proposed methodology around the two-stage learning/bridging pipeline.

Verification / evidence needed: The survey's architecture figure, abstract, gap, results and conclusion must all use the same story.

S-03 — P0 — Survey

Location: p.6, Fig. 1

Issue: Figure 1 places literature accuracy values (e.g. 98.43%) beside proposed AUC=0.743. These are different tasks and metrics.

Why it matters: It visually implies the proposed method is weaker than sentiment classifiers even though the research task is different.

Required correction: Replace with two panels or separate tables: literature sentiment-classification results vs launch-blind viability results.

Verification / evidence needed: Caption must explicitly state that metrics are not directly comparable.

S-04 — P0 — Survey

Location: p.13, Table 2

Issue: Proposed XGBoost row says “Target R^2>0.90” and MAE/RMSE “TBD.”

Why it matters: A submission cannot contain planned/TBD results in a final results table.

Required correction: Replace with final actual results, or remove the proposed row from that literature-comparison table.

Verification / evidence needed: Use the research paper's final external validation numbers in a separate launch-blind evaluation table.

S-05 — P0 — Survey

Location: References

Issue: The references are numbered with large gaps and citations are not in IEEE first-citation order. The first citation in the paper is [42].

Why it matters: IEEE numbered citations should be sequential by first appearance; the current numbering looks like an inherited/partially edited bibliography.

Required correction: Renumber all references after the content is frozen.

Verification / evidence needed: Do not manually patch individual numbers; regenerate the reference numbering globally.

S-06 — P0 — Survey

Location: p.4, p.9

Issue: Reference [31] is attributed to “A. Mori et al.” but arXiv:2407.13069 is by Junichiro Niimi.

Why it matters: This is a direct bibliographic error.

Required correction: Replace author/title metadata and verify the claimed 8.3% figure against the actual paper before retaining it.

Verification / evidence needed: If the exact 8.3% claim cannot be located in the source, delete the number or replace it with a verified result.

S-07 — P0 — Survey

Location: p.8, Section 4.1

Issue: Hidayat et al. [24] is described as an LLM→structured-feature→regressor architecture using narrative operational logs, but the cited IEEE Access paper is a time-series discrepancy forecasting study using Chronos and multiple forecasting models.

Why it matters: The cited source does not obviously support the architecture description currently assigned to it.

Required correction: Re-read the source and either correct the description or replace it with a paper that genuinely demonstrates LLM text-to-structured-feature regression.

Verification / evidence needed: Do not use it as the “closest architectural analog” unless the cited paper actually contains that architecture.

S-08 — P0 — Survey

Location: p.10–12

Issue: The survey says the proposed dataset includes Google Play apps, but the actual research paper is restricted to five physical Amazon product categories.

Why it matters: This makes the survey's methodology and claimed scope incompatible with the experiment.

Required correction: Remove Google Play from the final proposed-method dataset unless it is a separate experiment with results.

Verification / evidence needed: Keep the five final physical categories consistent across both papers.

S-09 — P1 — Survey

Location: p.1, p.7

Issue: “F1-Score is the most trusted evaluation metric” is interpretive language; the cited review supports frequency/importance, not a universal trust ranking.

Why it matters: It turns a descriptive literature statistic into an evaluative statement.

Required correction: Use “most frequently reported metric” or quote the exact wording of the systematic review.

Verification / evidence needed: Cite the Daza review directly.

S-10 — P1 — Survey

Location: p.5

Issue: The statement that SVM is unsuitable because it is “binary” is technically too broad. SVM can be extended to multiclass/multilabel settings with appropriate output schemes.

Why it matters: This is a factual simplification that a reviewer may catch.

Required correction: Say that a standard single SVM classifier does not directly produce a nine-dimensional structured aspect vector; multiple classifiers/output schemes would be required.

Verification / evidence needed: Avoid claiming that SVM itself is intrinsically binary-only.

S-11 — P1 — Survey

Location: p.3

Issue: The lexicon formula is presented as “the mathematical foundation” of lexicon-based sentiment scoring, but it is one possible TF-IDF-weighted formulation, not a universal canonical equation.

Why it matters: A survey should distinguish illustrative formulations from standard definitions.

Required correction: Label it “one illustrative weighted lexicon scoring formulation” or remove the equation.

Verification / evidence needed: Cite the method from the actual source if retained.

S-12 — P0 — Survey

Location: p.9

Issue: The retry formula uses p≈0.85 from another source and concludes P(valid|3 retries)≈0.997. The actual research system does not report an observed p=0.85 and the attempts are not necessarily independent.

Why it matters: The calculation is theoretical and the wording currently sounds like an empirical guarantee.

Required correction: Change “guarantee” to “theoretical probability under an independence assumption,” use “three attempts” rather than “three retries,” and preferably report actual observed validity rate from the 43k run.

Verification / evidence needed: If no measured p exists, remove the 0.997 figure from the survey.

S-13 — P1 — Survey

Location: p.10

Issue: The survey says the local Ollama/Gemma model was evaluated and excluded, while the synthesis later lists Ollama in the technology stack.

Why it matters: This creates an internal ambiguity about what is actually in production.

Required correction: Separate “evaluated but excluded” from “production stack.”

Verification / evidence needed: The final stack should match the research paper and deployed system exactly.

S-14 — P0 — Survey

Location: p.1, p.16

Issue: The claim “no published study has combined…” is too absolute.

Why it matters: A literature review can establish a gap within its search scope, not prove universal nonexistence.

Required correction: Use “within the studies identified by this review, we found no…” and state search databases/date/criteria.

Verification / evidence needed: Add the research paper's related prior-art distinction: Narayanan et al. pre-launch prediction uses pre-release e-WOM, whereas your setting uses specifications only.

S-15 — P1 — Survey

Location: p.16

Issue: The conclusion says “single-source, single-model systems consistently underperform hybrid systems.”

Why it matters: That is a broad comparative claim not demonstrated by the presented survey tables.

Required correction: Either provide a systematic quantitative comparison or soften to “hybrid systems can combine complementary signals.”

Verification / evidence needed: Do not infer universal superiority from heterogeneous studies.

S-16 — P1 — Survey

Location: p.8

Issue: Figure 2 is labelled “Illustrative SHAP feature-importance ranking” but the caption says LLM aspects dominate and “confirm” added signal.

Why it matters: An illustrative figure cannot serve as experimental evidence.

Required correction: Either generate the figure from the actual trained model or label it explicitly as a conceptual illustration and remove empirical language.

Verification / evidence needed: Use actual SHAP results from the research paper if available.

S-17 — P1 — Survey

Location: p.4, p.9

Issue: The survey uses the LLM retry/majority-voting literature to motivate the system, but the actual research implementation uses validation/retry across providers, not majority voting.

Why it matters: The mechanism should not be described as if it were the same method.

Required correction: Describe your actual retry/provider-fallback mechanism and cite majority voting only as related robustness literature.

Verification / evidence needed: Do not transfer a source's performance improvement to your own system.

S-18 — P1 — Survey

Location: p.11

Issue: The survey states zero-shot ABSA “eliminates” the need for labelled aspect data.

Why it matters: It removes the need for task-specific labels for the prompted extraction stage, but does not eliminate the need for validation or outcome labels for the downstream model.

Required correction: Use “reduces dependence on task-specific aspect labels for extraction.”

Verification / evidence needed: Keep the downstream product-level success labels clearly separate.

S-19 — P1 — Survey

Location: p.12

Issue: The survey says the aspects-only R^2=0.566 is the “leakage-honest headline” but the actual research paper's primary launch-blind evidence is the external holdout/temporal performance.

Why it matters: The survey's headline metric is not the same as the final research question.

Required correction: Make the external launch-blind holdouts the primary evidence and R^2 the intermediate-model diagnostic.

Verification / evidence needed: Align Abstract, Results and Conclusion.

S-20 — P1 — Survey

Location: p.1–17

Issue: The survey does not sufficiently explain the specification-to-aspect bridge, category grounding, trust tiers, abstention, or the ice-maker boundary condition.

Why it matters: These are now central to the actual research contribution.

Required correction: Add dedicated subsections: “Specification-to-Aspect Bridging,” “Launch-Blind Validation,” and “Boundary Conditions.”

Verification / evidence needed: Use the research paper's measured per-aspect fidelity and category results.

S-21 — P2 — Survey

Location: format

Issue: The document is A4, single-column, with mixed 9/10/12 pt text and many manually formatted sections; it is not safely treated as final IEEE conference formatting.

Why it matters: IEEE provides official templates and author tools; exact requirements depend on the target venue.

Required correction: Move the final text into the exact IEEE template for the target venue.

Verification / evidence needed: Do formatting only after technical content is frozen.

S-22 — P2 — Survey

Location: abstract

Issue: The survey abstract is about 300 words by the supplied document text.

Why it matters: IEEE conference guidance specifies a single-paragraph abstract up to 250 words.

Required correction: Reduce to ≤250 words and remove unnecessary implementation detail.

Verification / evidence needed: Use the final dataset and final launch-blind results.

6. Cross-paper consistency matrix — what must match exactly

Topic	Research paper final state	Survey must say	Status
Problem	Pre-launch / launch-blind product viability from specifications	Same	ALIGN
Historical data	43,000+ reviews; 640 initial products; five categories; 631 after filtering	Same	FIX SURVEY
Categories	Wireless headphones, Bluetooth speakers, smartwatches, power banks, ice makers	Same	FIX SURVEY
Target	Within-category composite of review volume, rating, velocity	Same, with explicit caveat that it is a proxy	ALIGN
Headline model	Leakage-prone rating/count/velocity removed; price retained; aspects + other pre-launch features	Same	FIX SURVEY
Bridge	Specification → estimated 9-aspect profile	Must be central	FIX SURVEY
Final external evidence	Holdout n=118; temporal n=125	Same exact numbers after recomputation	FIX BOTH
Holdout result	ρ=.529, AUC=.903 in text; figure conflicts	Use frozen recomputed values	FIX RESEARCH
Temporal result	ρ=.590, AUC=.942; within-cat ρ=.16	Same, with pooled vs within-cat distinction	ALIGN
Confidence	Reliability-weighted, but calibration not yet demonstrated	Describe as mechanism, not calibrated probability until tested	FIX BOTH
Trust tiers	Based on per-aspect bridge fidelity	Need threshold definition and training-only derivation	FIX RESEARCH
Boundary	Ice makers fail at bridging; personal electronics more expressive	Include in survey discussion	ADD SURVEY
SHAP	Relative product strengths/risks	Use actual results, not illustrative claims	FIX SURVEY
System	React + FastAPI + Docker; spec coach; analyzer	Only include features that are actually implemented and evaluated	VERIFY
Cold start	Specification-only inference, but trained on products that had reviews	Avoid absolute “cold-start” claims; use “launch-blind/low-information” carefully	FIX BOTH

7. Correct scientific narrative for both papers

The two papers should use the following exact conceptual chain. This is the cleanest version of your contribution:

```
Historical consumer evidence → 43,000+ reviews → null-forcing LLM ABSA → nine consumer aspects → XGBoost learns aspect-profile/viability relationship → trained viability model.

New product at launch → price + launch-available specifications + other verified pre-launch metadata → category-grounded specification-to-aspect bridge → estimated nine-aspect profile → trained XGBoost → viability/rank estimate → SHAP explanations + measured confidence + abstention.
```

This distinction fixes the leakage concern that appeared in the earlier survey version: the new product's review count, rating and review velocity are not used as predictive inputs at inference time. They are part of the historical outcome used to teach the model what “viability” means.

Important wording: the system is not predicting sales and is not predicting absolute failure for products with no market attention. The validated claim is a survivor-product, within-category viability ranking from launch-available specifications, with strong variation by category.

8. Figure and table audit

Paper	Current figure/table	Problem	Correction
Research	Fig. 1	Chart text says n=732; final dataset is 631. Also mixes regression R^2 with classification metrics on one scale.	Regenerate with n=631; separate regression and top/bottom-quintile classification panels.
Research	Fig. 2	Runtime/accuracy language is ambiguous; R^2 is not "accuracy."	Rename to predictive performance and training efficiency; report benchmark environment.
Research	Fig. 3	n=732 in artwork; "BEST/Good/Weak" overall verdict unsupported by all metrics.	Remove verdict; use bold per-metric best values only.
Research	Fig. 4	Confusion-matrix totals/protocol need verification; labels say flops/hits.	Recompute from out-of-fold predictions and use bottom/top quintile labels.
Research	Fig. 5	Architecture is good but needs explicit training-only category-profile boundary.	Add a visual "TRAINING ONLY" wall around review-derived category statistics.
Research	Fig. 6	Numerical annotation conflicts with text/Table 2.	Regenerate from frozen holdout predictions.
Research	Fig. 7	Useful, but trust-tier thresholds and confidence interpretation are not defined.	Show exact CI method and define tier thresholds in Method; ideally validate thresholds.
Research	Fig. 8	Good application evidence, but it is not itself model validation.	Keep as system demonstration; do not treat UI as experimental evidence.
Survey	Fig. 1	Accuracy vs AUC from different tasks is visually misleading; proposed result is outdated.	Replace with literature landscape + separate launch-blind result panel.
Survey	Fig. 2	Illustrative SHAP plot is described as confirming a result.	Use actual SHAP or mark conceptual and remove evidence claim.
Survey	Fig. 3	Should show the two-stage bridge architecture, not only review→LLM→XGBoost.	Redraw to match research paper.
Survey	Fig. 4	Theoretical retry curve uses an unverified $p \approx .85$ as if it were system reliability.	Replace with observed validity curve or remove.
Survey	Fig. 5	Label weights figure is okay, but it should be clearly identified as target-label design, not learned importance.	Retain with sensitivity-analysis discussion.
Survey	Fig. 6	Algorithm-frequency chart is useful but should not imply performance superiority.	Keep as literature-frequency evidence, not model-selection evidence.
Survey	Tables 2–3	Mixed task metrics and TBD proposed values.	Separate literature sentiment classification from your launch-blind viability evaluation.

9. Minor but important scientific wording corrections

Current style	Use instead	Reason
"F1 is the most trusted metric"	"F1 was the most frequently reported metric in the reviewed studies."	Descriptive, not evaluative.
"no published study has…"	"Within the studies identified by our review, we found no study that…"	Limits the claim to the search scope.
"SVM is binary-only"	"A standard SVM classifier does not directly output a nine-dimensional aspect vector without multiple output models."	Technically precise.
"P(valid)=0.997 guarantee"	"Under an independence assumption and p=..., the theoretical probability after three attempts is..."	Not an empirical guarantee.
"viability falls as price rises"	"The headline model is constrained so predicted viability cannot increase with price when other inputs are held fixed."	Distinguishes imposed monotonicity from causal effect.
"orthogonal to price"	"weakly correlated with price (r=0.147)"	Orthogonality is stronger terminology.
"flops/hits"	"bottom-quintile/top-quintile survivors"	Matches the target definition.
"only honest thing"	"the system exposes measured uncertainty and abstention conditions."	Scientific tone.
"default to beat"	"strong baseline for tabular prediction"	Scientific tone.
"answers the most common objection"	"provides a temporal generalization check."	Avoids unsupported universal claim.

10. Literature/reference audit

A limited external verification was performed for selected high-risk citations. The point is not to replace a full reference-manager check, but to identify citations that should be corrected before submission.

Item	Finding	Action
Daza et al. 2024	The published article is IJIM Data Insights 4(2), article 100267, DOI 10.1016/j.jjimei.2024.100267; it reviews 20 studies across four databases.	Keep, but use exact metadata and describe its findings as reported frequencies/summary results.
Shan et al. 2025	IEEE Access 13, 51027–51041, DOI 10.1109/ACCESS.2025.3551666. The source supports ELECTRA 98.09%, BERT 95.28%, BiLSTM 96.09%.	Correct survey reference title to the exact published title and retain task-specific context.
Niimi / arXiv:2407.13069	The cited “Mori et al.” source is actually Junichiro Niimi, “Dynamic Sentiment Analysis with Local Large Language Models using Majority Voting...”.	Fix author metadata and verify the claimed 8.3% number before using it.
Elangovan & Subedha 2023	Their paper reports XGBoost 91.60% as a comparison baseline on the CPAA dataset; their proposed APGWO-DLSA is 94.77%.	Survey must not imply that XGBoost was their proposed method.
Thomas & Jeba 2024	Publisher page reports 98.43% accuracy, but currently displays an investigation notice.	Use cautiously; verify current status and avoid making this paper a cornerstone of the survey's argument.
Hou et al. 2024	The paper introduces Amazon Reviews 2023; it describes a much larger source dataset than your filtered 43k-review sample.	Cite it as the dataset source, then document your exact filtering/sampling pipeline separately.
Hidayat et al. 2025	The cited IEEE Access work is a Chronos/time-series discrepancy forecasting study.	Do not describe it as an LLM narrative-to-regressor ABSA analogue unless the actual source supports that description.

Survey reference-numbering problem

The survey currently contains references such as [1]–[9], then [15], [16], [17], [18], [20], [24], [25], [31], [32], [35], [39]–[47]. The body also cites [42] before many lower numbers. This is not a clean IEEE numbered-reference sequence. After the content is finalized, renumber references by first appearance and regenerate all in-text citations automatically. Do not preserve the existing inherited numbers just to avoid editing the text.

11. IEEE-format audit

Requirement/check	Research paper	Survey paper	Action
Official IEEE template	Custom layout	A4 single-column Word layout	Use exact target IEEE template.
Abstract length	~343 words	~300 words	Reduce to ≤250 words.
Abstract style	Contains rhetorical/strong claims	Dense and method-heavy	One paragraph; self-contained; no references/equations.
Author block	Only Raj Jagtap + explanatory note	Multiple authors + affiliation	Use target venue's exact author/affiliation format.
Citations	Author-year	Numbered but not IEEE order	Convert both to venue-required IEEE numeric style if conference/journal requires it.
References	Alphabetical author-year style	Numbered, but inconsistent sequence	Regenerate with IEEE bibliography style.
Figures/tables	Mostly good but stale numbers and unsupported verdicts	Several illustrative figures presented too strongly	Regenerate after analysis freeze.
Page count	21 pages	17 pages	Check the exact target venue limit; do not assume a generic IEEE limit.

IEEE's Author Center states that conference abstracts should be a single paragraph up to 250 words, self-contained, with no references, footnotes, or mathematical equations, and recommends using the official IEEE templates and author tools. Exact page limits and formatting details remain venue-specific.

12. Recommended final revision sequence

Step	Do this	Why first?
1	Freeze the actual dataset: 640 initial → 9 fraud-filter removals → 631 analysis products; document every exclusion.	Prevents every downstream figure/table from drifting.
2	Freeze the target definition, normalization, velocity window, and label-weight sensitivity protocol.	Defines what “viability” actually means.
3	Freeze feature provenance: specs, price, brand tier, category profile, pain points, calibration reference, trust weights.	Closes hidden leakage paths.
4	Recompute the 118-product holdout from scratch and regenerate Figure 6.	Resolves the largest numerical contradiction.
5	Recompute the 125-product temporal holdout and explain the missing 631st product.	Makes the temporal experiment auditable.
6	Recompute top/bottom quintile AUC/confusion matrices from clearly defined labels.	Fixes the classification diagnostic.
7	Run rating-blind ABSA extraction ablation.	Tests hidden target-proxy leakage.
8	Run constrained vs unconstrained model ablation.	Supports the monotonic-constraint claim.
9	Run per-category and macro-average evaluation with CIs.	Prevents pooled metrics from hiding category failure.
10	Run confidence/abstention calibration and risk–coverage analysis.	Validates the “honesty layer.”
11	Update research paper text/figures/tables/references.	Research paper becomes the final source of truth.
12	Rewrite survey around the frozen research paper.	Prevents the two papers diverging again.
13	Convert both into the exact IEEE template and regenerate references automatically.	Formatting should be last, not first.
14	Run a final consistency script/search for all key numbers and terms.	Catches stale 732/631, .437/.529, old AUC .743, Google Play, TBD, etc.

13. Final pre-submission search list

Run a literal text search across both final manuscripts for the following terms/numbers and confirm every occurrence is intentional:

```
Numbers: 43,600, 43,000, 640, 631, 732, 1,051, 74,000, Google Play, 0.437, 0.529, 0.903, 0.826, 0.590, 0.942, 0.743, 0.595, 0.566

Claims/terms: 0.892, TBD, Target R², free-tier, Ollama, Mori, Hidayat, flop, hit, cold-start, no published, only honest, default to beat, orthogonal, 98.43%, 91.60%
```

14. Submission-readiness verdict

Paper	If submitted now	After required corrections
Research paper	**Not ready**: strong core, but numerical/figure inconsistencies and protocol gaps are too visible.	Potentially strong IEEE-style experimental paper if the recomputation confirms the headline results and the provenance/ablation/calibration issues are addressed.
Survey paper	**Not ready**: methodology, dataset, figure comparison, references and proposed results are out of sync with the research paper.	Good companion survey/framework paper after it is rewritten around the final launch-blind architecture and literature scope.

Bottom line

The central research idea is worth keeping. The biggest risk is not that the model is “too low accuracy”; it is that the paper currently contains several places where a reviewer can no longer tell which experiment generated which number. Fixing provenance and evaluation consistency is therefore more important than trying to increase a single headline metric. In particular, do not optimize the paper around the 0.590/0.942 pooled temporal result while hiding the 0.16 within-category temporal correlation. Present both: the pooled signal and the conservative within-category generalization result. That is scientifically stronger and much harder to criticize.

15. External verification notes

Selected external checks used for this audit:

- IEEE Author Center — Structure Your Paper: conference abstract guidance (single paragraph, up to 250 words, self-contained).
- IEEE Author Center — Authoring Tools and Templates: official IEEE templates and reference-validation tools.

- Daza et al. (2024), International Journal of Information Management Data Insights, 4(2), 100267, DOI 10.1016/j.jjimei.2024.100267.
- Shan, Sun & Macawile (2025), IEEE Access 13, 51027–51041, DOI 10.1109/ACCESS.2025.3551666.
- Niimi (2024), arXiv:2407.13069 — the source currently attributed in the survey to “Mori et al.”
- Elangovan & Subedha (2023), Engineering, Technology & Applied Science Research 13(3), 10989–10993, DOI 10.48084/etasr.5787.
- Hou et al. (2024), “Bridging Language and Items for Retrieval and Recommendation,” arXiv:2403.03952 — source describing Amazon Reviews 2023.

This report is an audit and correction plan, not a claim that the proposed model is valid merely because the current numbers look promising. The decisive next step is to regenerate the affected analyses from the final frozen data/prediction artifacts and make both papers report those same verified outputs.